

Research on a Spectral Reconstruction Method with Noise Tolerance

Yunlong Ye, Jianqi Zhang, Delian Liu*, and Yixin Yang

School of Physics and Optoelectronic Engineering, Xidian University, Xi'an 710071, China

(Received May 24, 2021 : revised July 18, 2021 : accepted July 26, 2021)

As a new type of spectrometer, that based on filters with different transmittance features attracts a lot of attention for its advantages such as small-size, low cost, and simple optical structure. It uses post-processing algorithms to achieve target spectrum reconstruction; therefore, the performance of the spectrometer is severely affected by noise. The influence of noise on the spectral reconstruction results is studied in this paper, and suggestions for solving the spectral reconstruction problem under noisy conditions are given. We first list different spectral reconstruction methods, and through simulations demonstrate that these methods show unsatisfactory performance under noisy conditions. Then we propose to apply the gradient projection for sparse reconstruction (GRSR) algorithm to the spectral reconstruction method. Simulation results show that the proposed method can significantly reduce the influence of noise on the spectral reconstruction process. Meanwhile, the accuracy of the spectral reconstruction results is dramatically improved. Therefore, the practicality of the filter-based spectrometer will be enhanced.

Keywords : Compressive sensing, Noise tolerance, Spectrometer

OCIS codes : (300.6190) Spectrometers; (300.6550) Spectroscopy, visible

I. INTRODUCTION

Spectrum analysis is an important technology in almost all fields of science [1–4]. As an important tool to separate and measure a spectrum, the spectrometer attracts more and more attention with the development of spectroscopy technology. Current spectrometers, such as the prism, grating, and interferometer types, are bulky, expensive, and delicate, which greatly limits their applications [5]. The mechanisms of these spectrometers are shown in Fig. 1. A prism instrument uses the difference in refractive index between different wavelengths of light to separate optical bands [Fig. 1(a)] [6]. However, an additional scanning mechanism is required to obtain the spectral bands. This increases the volume of the system, and accuracy is difficult to guarantee. A grating instrument uses the diffraction of light to separate optical bands [Fig. 1(b)] [7, 8]. The grating will attenuate the light intensity, and is difficult to fabricate and calibrate.

An interferometer uses the interference of light to separate optical bands [Fig. 1(c)] [9]. The structure of the interferometer is complex, making it huge and expensive. To make instruments with better performance and a wider range of applications, a lot of effort has been invested to design small, inexpensive and simple spectrometers [10–12].

Recently, the seminal work of Bao and Bawendi [13] has shown a new type of spectrometer, based on a set of filters whose transmittance features can be varied. The mechanism of this filter-based spectrometer is shown in Fig. 1(d). The original light first passes through a given filter of predetermined transmittance; then the radiance of the transmitted light is measured by a detector. This radiance measurement needs to be repeated many times, using different filters with different transmittances. It is efficient to place these filters and their corresponding detectors in an array format. The spectrum of the original light can be reconstructed from these radiance values, using a spectral reconstruction method.

*Corresponding author: dliliu@xidian.edu.cn, ORCID 0000-0001-8655-8010

Color versions of one or more of the figures in this paper are available online.



This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/4.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

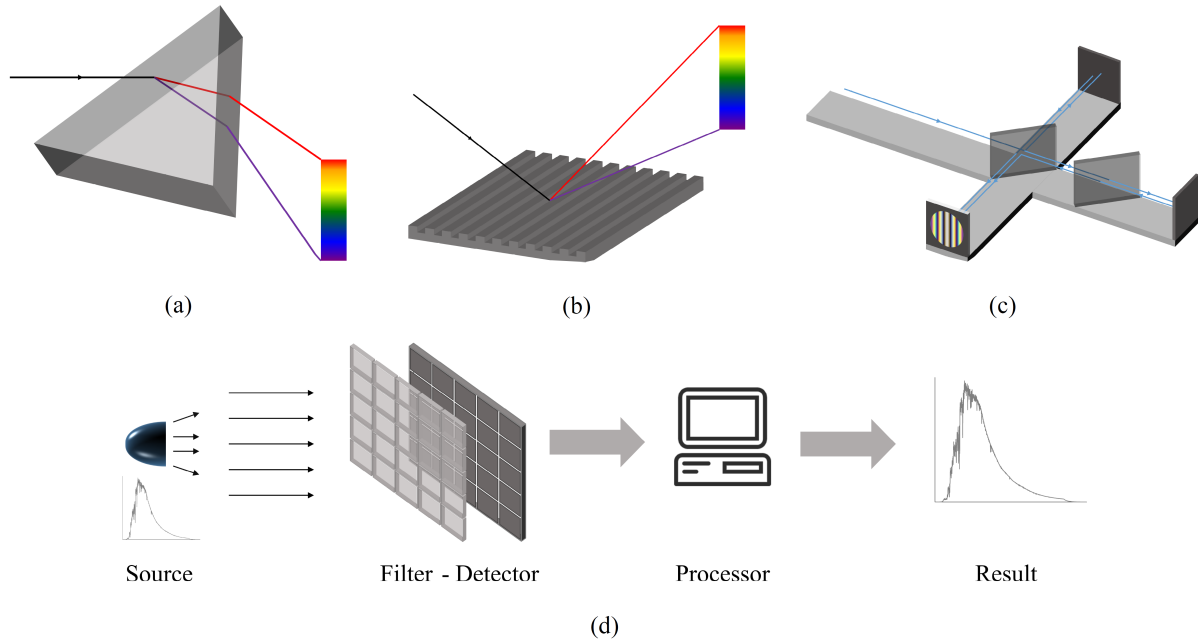


FIG. 1. Mechanisms of different spectrometers: (a) based on a prism, (b) based on a grating, (c) based on an interferometer, and (d) based on filters with different transmittance features.

Since filter-based spectrometers can greatly reduce size and cost, many efforts have been made to propose new filters and various spectral reconstruction methods recently.

In the research of Bao and Bawendi [13], the filter array is composed of colloidal quantum dots (CQDs), and the transmittance of the filter can be changed simply by changing the size, shape, and composition of the CQD. The research mainly solves the problem when the number of filters is larger than the number of spectral bands. In such a case, the reconstruction method uses the least-squares (LS) linear regression algorithm. All of the major features of the spectrum can be reconstructed and the spectral range can be increased simply by increasing the number of filters.

Yang *et al.* [14] proposed an approach based on a single compositionally engineered nanowire. By measuring the photocurrent along the length of the nanowire, the spectrum can be reconstructed. The study introduced an adaptive Tikhonov regularization algorithm to the spectral reconstruction method, to reduce the influence of the system's noise, which can make the reconstruction more accurate.

Wang and Yu [15] from the University of Wisconsin, Madison proposed to apply a compressive sensing algorithm to the spectral reconstruction method, which can solve the problem when the number of filters is less than the number of spectral bands. In their research, the filter array is composed of different nanophotonic structures. The spectrum of the source light is reconstructed by the basis pursuit (BP) algorithm, and the spectral position and radiance of the original light can be recovered very well.

When spectrometers are applied under practical conditions, many additional factors such as noise need to be considered. Noise can be generated in every step of the spectral

reconstruction process [16], and it can greatly interfere with the accuracy of the reconstruction. Therefore, it is worthwhile to study denoising approaches. The spectrometers we have introduced above can perform well under their applicable conditions, but their reconstruction performance in the face of noise is unsatisfactory. One of the important factors affecting denoising is the spectral reconstruction method. The reconstruction method in [13] used the LS algorithm; while it can perform well under ideal conditions, it is unable to reconstruct the spectrum of the original light under noise. The reconstruction method in [14] was based on an adaptive Tikhonov regularization algorithm, which can offset the deficiency of the LS algorithm in noise processing. However, the denoising ability of this method still needs to be improved. The method in [15] used the BP algorithm; it was proposed to solve the problem when the number of filters is less than the number of spectral bands, which allows the reconstruction method to have a wider operational range with fewer filters. However, the BP algorithm is not designed for recovering a signal under noisy conditions, so the method based on the BP algorithm can only work when noise is not a concern.

In this paper, we focus on denoising approaches for the filter-based spectrometer. After comparing several methods through simulations, we analyze the effect of noise on the spectral reconstruction performance, and to improve the reconstruction effect we propose a new spectral reconstruction method. The gradient projection for sparse reconstruction (GPSR) algorithm is used to solve the convex unconstrained optimization problem; the algorithm shows good accuracy, so it is used in many fields for signal reconstruction [17, 18]. The GPSR algorithm has not been introduced

for spectral reconstruction in such filter-based spectrometers in the literature, and we propose applying the GPSR algorithm to their spectral reconstruction method. In addition, we verify the reconstruction effect of this method through simulations. The results show that the spectral reconstruction method proposed in this paper has good performance in recovering the spectrum under noise coming from the system, and the method can achieve good reconstruction performance with fewer filters.

Our work can make filter-based spectrometers more practical, and the main contributions of this paper are summarized as follows:

- (1) The effect of noise on the spectral reconstruction is analyzed.
- (2) Different spectral reconstruction methods are analyzed, to compare their noise tolerance.
- (3) The spectral reconstruction method based on the GPSR algorithm is proposed, to improve the denoising effect of the spectrometer.
- (4) Simulations are designed to verify the effects of the denoising approaches.

II. ANALYSIS OF THE NOISE EFFECT

In [13], [14], and [15], it has been proved that it is possible to realize spectral measurement by passing the source light through a set of filters with different spectral response characteristics, and in this section we show the effect of noise on the spectral reconstruction performance.

The spectral reconstruction process for a filter-based spectrometer can be described mathematically as shown in Fig. 2. In the ideal case, we consider an unknown source light $\Phi(\lambda)$ (λ is the wavelength) passing through a two-dimensional filter array, where the transmittance of filter i can be described as $T_i(\lambda)$ ($i = 1, 2, 3, \dots, n_F$, n_F is the number of filters). The transmitted light through each filter can be measured by the corresponding detector, and the output value L_i ($i = 1, 2, 3, \dots, n_F$) of the i^{th} detector is in accordance with Eq. (1):

$$L_i = \int \Phi(\lambda) T_i(\lambda) K(\lambda) d\lambda \quad (i = 1, 2, 3, \dots, n_F). \quad (1)$$

In Eq. (1), $K(\lambda)$ is the response function of the detector, and we regard $K(\lambda)$ of each detector to be the same. Therefore, we ignore the influence of $K(\lambda)$ and assume that the response function of the detector $[K(\lambda)]$ at any wavelength is 1 in this paper. In the actual spectral reconstruction process, the variables in Eq. (1) can only be in the discrete form, and Eq. (1) can be discretized into a form such as

$$L_i = \sum_{\lambda} T_i(\lambda) \Phi(\lambda) \quad (i = 1, 2, 3, \dots, n_F). \quad (2)$$

In Eq. (2), the wavelength λ is discretized into n_{λ} sections, $T_i(\lambda)$ is predetermined, and L_i can be measured by the detector; only $\Phi(\lambda)$ is unknown, so the spectrum of the original light can be reconstructed by finding the solution of Eq. (2). Equation (2) can be regarded as a set of linear equations with n_{λ} unknowns and n_F equations, and different spectral reconstruction methods should be applied in different cases to recover the original signal $\Phi(\lambda)$. In the ideal case of $n_{\lambda} = n_F$, Eq. (2) is well-posed and the equations have a unique solution; when $n_{\lambda} < n_F$, Eq. (2) is overdetermined and has a least-squares solution; and when $n_{\lambda} > n_F$, Eq. (2) is underdetermined and the solution of the linear equations is uncertain.

Equation (1) shows the spectral reconstruction process in the ideal situation; when we carry out a physical experiment or apply the method to an actual situation, noise must be a concern. Noise is the useless information in the spectral reconstruction system, and its existence greatly affects the result of the spectral reconstruction [16, 19]. In a spectrometer, noise exists at every step, such as the unwanted signal due to detector noise and stray light, the measurement error generated during the measurement of the filter's transmittance, and the drift of transmittance induced by temperature change. Different types of noise have different effects on spectral reconstruction [20]. To describe the system's noise and the modeling mismatch [21, 22], a noise δ_i is added to Eq. (2):

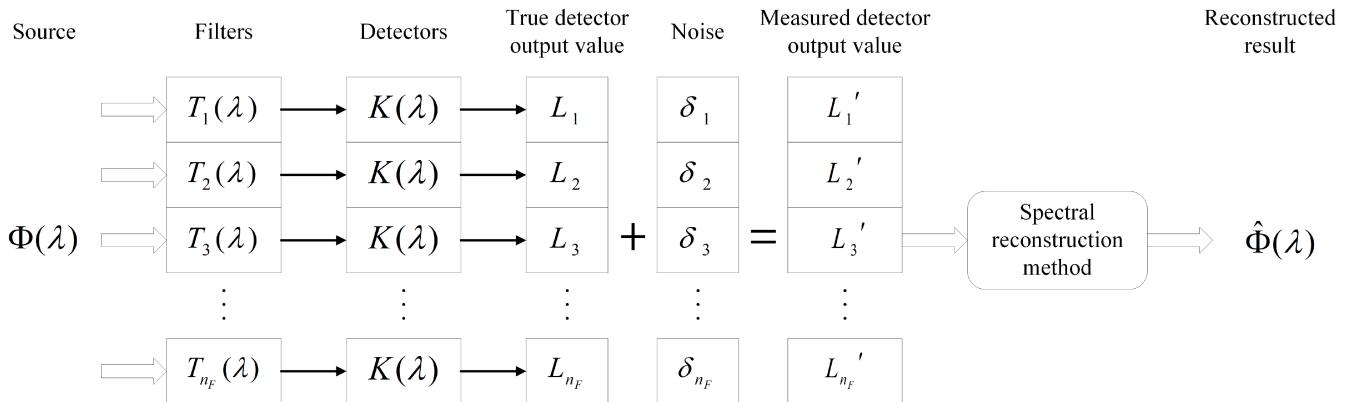


FIG. 2. Schematic of the spectral reconstruction process.

$$L'_i = L_i + \delta_i = \sum_{\lambda} T_i(\lambda) \Phi(\lambda) + \delta_i \quad (i=1, 2, 3, \dots, n_F), \quad (3)$$

where L_i is the true value of the transmitted light radiance, and L'_i is the measured value of the transmitted light radiance obtained by the detector. The noise tolerance of different algorithms can be different; to recover the original signal $\Phi(\lambda)$ as accurately as possible, a signal recovery algorithm with better noise tolerance should be chosen.

According to previous research, spectral reconstruction methods based on various algorithms have been applied, such as the LS algorithm, the BP algorithm, or the Tikhonov regularization algorithm, for example. In our work, we analyze the spectral reconstruction performance of these methods under noise through simulated experiments.

We design the simulations to evaluate the effect of the noise on the spectral reconstruction results, and the noise tolerance of the different methods mentioned above. The comparison of the performance of different spectral reconstruction methods shows that noise has a great influence on the reconstruction.

The process of spectral reconstruction is simulated in MATLAB. We consider the source light $\Phi(\lambda)$ with the wavelength λ ranging from 400 to 800 nm; $\Phi(\lambda)$ contains n_{λ} unknowns, so the resolution is $(800 - 400) / (n_{\lambda} - 1)$ nm. To simplify the simulation process, we assume that the radiance of the source light ranges from 0 to 1. As shown in Figs. 3 (a) and 3(b), we design a single-spectral-peak light signal and a broadband light signal.

Then we design n_F filters $\mathbf{T} = [T_1(\lambda), T_2(\lambda), T_3(\lambda), \dots, T_{n_F}(\lambda)]^T$, where a filter consists of multiple layers, and different layers have staggered grating structures, such that the transmittance of the filter can be easily adjusted by changing the period of the grating. We designed 100 sets of filters of different transmittance; transmittance traces for some of the filters are shown in Fig. 3(c).

To evaluate the effect of noise on the spectral reconstruction, we compare the reconstruction results for different methods, in both ideal and noisy cases. After obtaining the

reconstruction results based on different algorithms, we use the root mean square error (RMSE) to evaluate the quality of the reconstruction results:

$$\text{RMSE}[\Phi'(\lambda), \Phi(\lambda)] = \sqrt{\frac{[\Phi'(\lambda) - \Phi(\lambda)]^2}{n_{\lambda}}}. \quad (4)$$

RMSE is widely used to measure the deviation between observations and true values. In our research we calculate the RMSE for different spectral reconstruction methods under the same simulation conditions and noise levels. The smaller the RMSE, the closer the reconstructed signal is to the original signal, which means better reconstruction performance.

2.1. The Least-squares Linear Regression Algorithm

The least-squares linear regression algorithm has been widely used to reconstruct spectra in previous research [13, 23, 24]. When we reconstruct the original spectrum in the ideal case, Eq. (2) can be converted to a least-squares (LS) problem:

$$\min_{\Phi} \|\mathbf{L} - \mathbf{T}\Phi\|_2^2. \quad (5)$$

Research on the least-squares problem is quite mature. The LS algorithm is not proper for an underdetermined system, so the algorithm cannot solve the spectral reconstruction problem when the number of filters n_F less than the number of spectral bands n_{λ} . Therefore, to achieve higher resolution in spectral reconstruction, more filters are needed, which greatly increase the volume of a filter-based spectrometer.

To evaluate the effect of system's noise on a method based on the LS algorithm, we generate random white Gaussian noise $\mathbf{\Delta} = (\delta_1, \delta_2, \delta_3, \dots, \delta_{n_F})^T$ as the system's noise in Eq. (3). In this paper, we use S/N to describe the level of noise, it is defined as the ratio of the root mean square amplitude of the signal to the noise. After we obtain the measurement of radiance $\mathbf{L}'$, the original spectrum Φ can

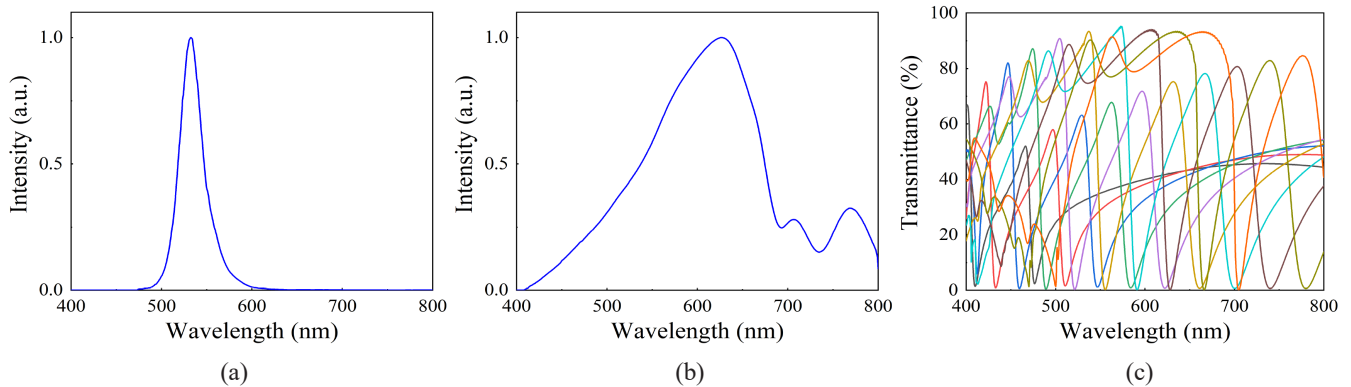


FIG. 3. Simulation of the source light and the filters: (a) the single-spectral-peak light signal, (b) the broadband light signal, and (c) the transmission fraction for some of the filters.

be reconstructed using the LS algorithm. The reconstruction results are shown in Figs. 4(a) and 4(b); we obtain the reconstruction results for a single-spectral-peak light signal and a broadband light signal respectively. The RMSE of the reconstructed signal in Fig. 4(a) is 1.054×10^2 , while that of the signal in Fig. 4(b) is 1.564×10^3 . The reconstruction results show that the method based on the LS algorithm cannot reconstruct the spectrum of the original signal from noisy observations. The explanation of this phenomenon is as follows:

Although the transmittances of different filters are different, there are still some correlations between them. This makes the system of equations ill-conditioned, which means that a very small disturbance in $\mathbf{T}$ and $\mathbf{L}$ in Eq. (2) will have a huge influence on the reconstruction results. Therefore, when the LS algorithm is used for spectral reconstruction, the presence of noise will make the reconstruction results deviate strongly from the original spectral signal.

2.2. The BP Algorithm

In Wang and Yu's paper [15], they applied the BP algorithm in the spectral reconstruction method. This method can use fewer filters while meeting a certain reconstruction spectral resolution, which can further reduce the volume of the spectrometer. In this paper, we mainly deal with the case when $n_\lambda > n_F$.

The compressive sensing algorithm [25–28] is an effective way to solve the problem when $n_\lambda > n_F$. The method of compressive sensing is shown in Fig. 5. A signal $\mathbf{x}$ (one-dimensional signal of length n) is a *sparse* signal when $\mathbf{x}$ only has k nonzero coefficients ($k \ll n$). When the signal $\mathbf{x}$ is sparse or approximately sparse, we can define a measurement matrix $\mathbf{T}$ to project the high-dimensional signal $\mathbf{x}$ onto the low-dimensional space, and the projection process can

be described as

$$\mathbf{y} = \mathbf{T}\mathbf{x}. \quad (6)$$

The measurement $\mathbf{y}$ obtained in this way is a one-dimensional signal of length $m(m < n)$, and $\mathbf{y}$ contains all of the information of the original signal $\mathbf{x}$. If the signal $\mathbf{x}$ is not sparse, then a sparsifying basis Ψ (matrix of size n by n) can be found such that the signal $\mathbf{x}$ can be sparse or approximately sparse in Ψ ; that is,

$$\mathbf{x} = \Psi\boldsymbol{\theta}, \quad (7)$$

where $\boldsymbol{\theta}$ is a sparse or approximately sparse signal. Then the projection process shown in Eq. (6) can be rewritten as

$$\mathbf{y} = \mathbf{T}\mathbf{x} = \mathbf{T}\Psi\boldsymbol{\theta} = \mathbf{\Theta}\boldsymbol{\theta}, \quad (8)$$

where $\mathbf{\Theta} = \mathbf{T}\Psi$. When we get the measurement $\mathbf{y}$, the sparse signal $\hat{\boldsymbol{\theta}}$ can be obtained by solving $\mathbf{y} = \mathbf{\Theta}\boldsymbol{\theta}$, and then the reconstruction result $\hat{\mathbf{x}}$ can be obtained through Eq. (7).

The spectral reconstruction process is similar to the compressive sensing process, in which the unknown source light $\Phi(\lambda)$ can be regarded as the signal $\mathbf{x}$ that will be solved, the filter response function $\mathbf{T}$ is equivalent to the measurement matrix $\mathbf{T}$, and the radiance of the transmitted light $\mathbf{L}$ is equivalent to the measurement $\mathbf{y}$. Because of the similarity between the spectral reconstruction process and the compressive sensing process, the solution $\hat{\Phi}(\lambda)$ of Eq. (2) can be found by using the relevant algorithm of compressive sensing in the case when $n_\lambda > n_F$.

A typical compressive sensing algorithm that can be used to solve the spectral reconstruction problem is the BP algorithm [29], the main idea of which is to convert the

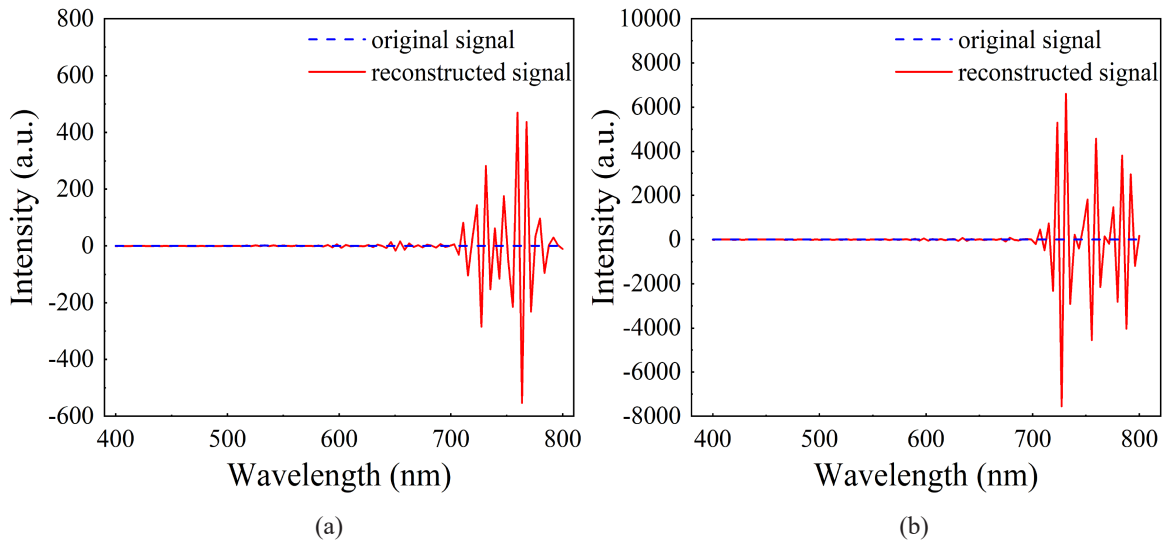


FIG. 4. The effect of system's noise on the LS algorithm: (a) the reconstruction results for a single-spectral-peak light signal where noise is concerned, with $S/N = 1000/1$ and $n_\lambda = n_F = 100$, and (b) the reconstruction results for a broadband light signal where noise is concerned, also with $S/N = 1000/1$, and $n_\lambda = n_F = 100$.

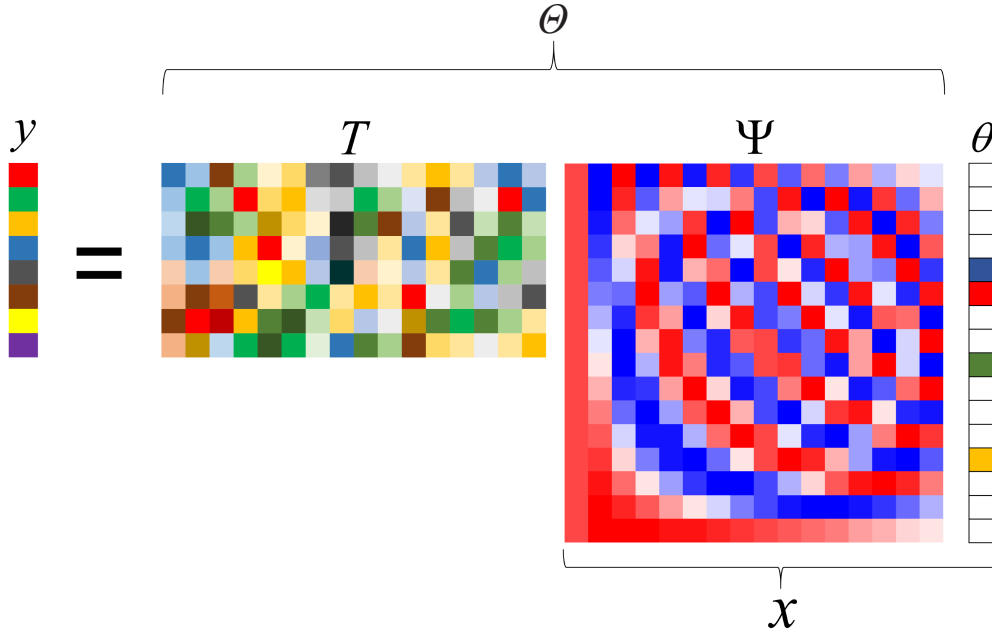


FIG. 5. The method of compressive sensing.

underdetermined linear problem into a convex optimization problem:

$$\min_{\theta} \|\theta\|_1 \quad \text{subject to } \Theta \theta = L, \quad (9)$$

where $\Theta = T\Psi$. The method based on the BP algorithm has satisfactory performance in the noiseless case, but the BP algorithm is not designed for recovering a signal when noise is concerned.

The effect of noise on the reconstruction performance can be evaluated through the above simulated experiments. We compare the spectral reconstruction performance in the ideal case and under noise, and again select two light sources, the single-spectral-peak light source and the broadband light source. For a non-sparse signal, the sparsifying basis selected is the discrete cosine transform (DCT) matrix. The method based on the BP algorithm cannot work under noise, so to reconstruct the spectrum from noisy observations more accurately, a method with better noise tolerance should be found.

As shown in Figs. 6(a) and 6(c), the RMSE of the reconstructed signal in Fig. 6(a) is 1.83×10^{-3} , and that in Fig. 6(c) is 2.43×10^{-3} . Obviously, the reconstruction results almost coincide with the original spectral signal; the method based on the BP algorithm can solve the underdetermined linear problem, so the method can deal with the case when $n_\lambda > n_F$, when there is no noise. The spectral reconstruction results when noise is concerned are shown in Figs. 6(b) and 6(d), for which the RMSE is 0.414 and 3.010 respectively. From the reconstruction results, we can see that the method based on the BP algorithm dose not offer satisfactory performance under noise.

2.3. The Tikhonov Regularization Algorithm

The spectral reconstruction method in [14] uses the Tikhonov regularization algorithm. As mentioned, when system's noise exists in the spectral reconstruction process, the reconstruction performance of traditional algorithms without noise tolerance is greatly affected. One way to reduce the influence of system's noise is to add a regularization term to the quadratic minimization problem [30]. The method based on the Tikhonov regularization algorithm can solve the ill-posed problem in linear equations, so it has the feature of noise tolerance.

The Tikhonov regularization algorithm obtains the solution of Eq. (3) by finding the minimum point of the regularization problem:

$$\min_{\theta} \|\mathbf{L}' - \mathbf{T}\Phi\|_2^2 + \tau \|\theta\|_2^2, \quad (10)$$

where $\tau > 0$ is the regularization parameter. We use the same simulations as above to evaluate the reconstruction performance of the method based on the Tikhonov regularization algorithm under noise.

As shown in Figs. 7(a) and 7(c), the RMSE of the reconstructed signal in Fig. 7(a) is 1.49×10^{-3} and that in Fig. 7(c) is 2.39×10^{-3} . Therefore, when the noise is not considered, the method based on the Tikhonov regularization algorithm can reconstruct the spectrum when $n_\lambda > n_F$, and the spectral reconstruction performance of the method is nearly perfect. When noise is concerned, the spectral reconstruction results are shown in Figs. 7(b) and 7(d), where the RMSE of the reconstructed signal in Fig. 7(b) is 5.29×10^{-3} and that in Fig. 7(d) is 2.26×10^{-2} .

The spectral reconstruction results show that the method

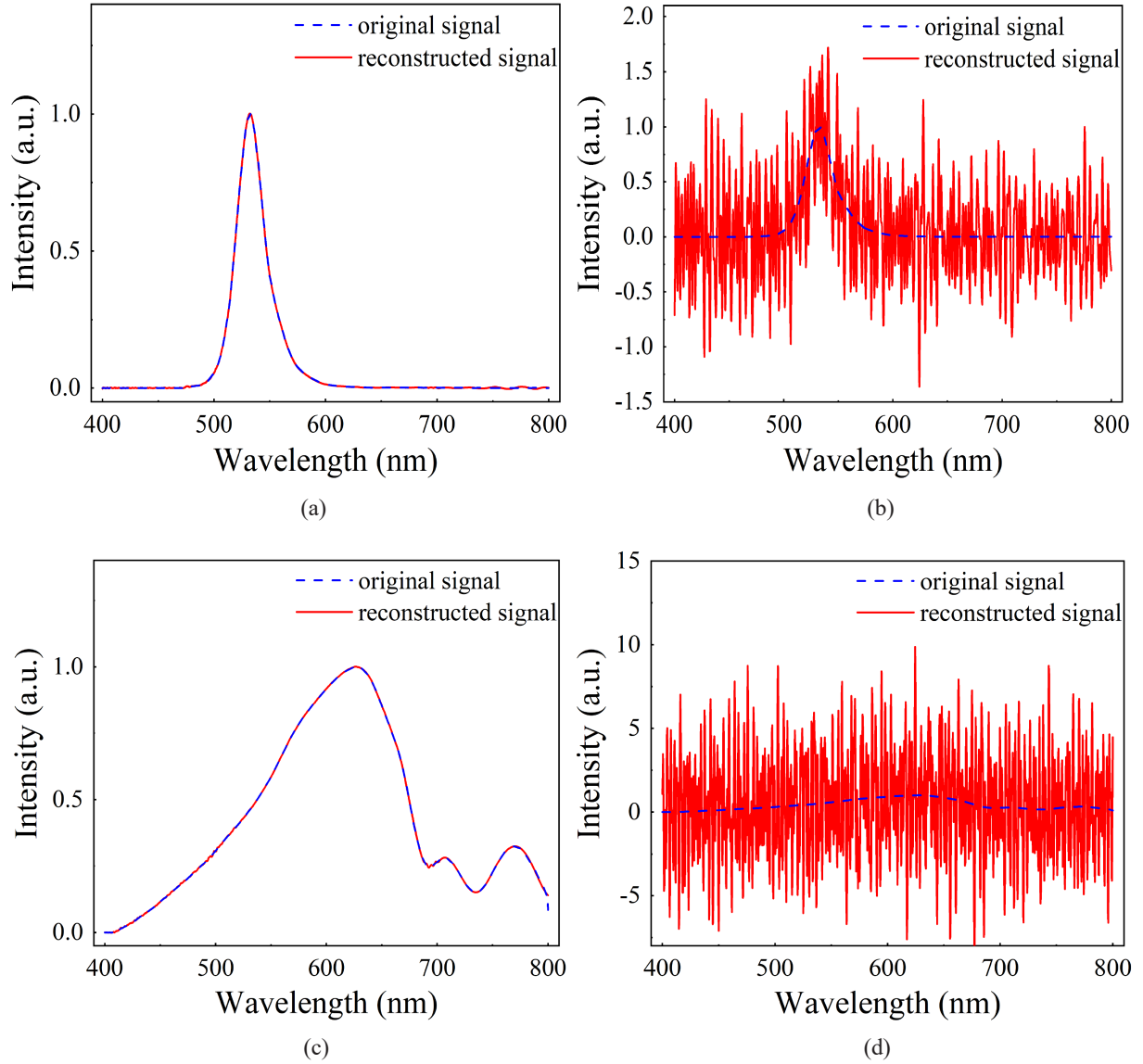


FIG. 6. The effect of system's noise on the basis pursuit (BP) algorithm. (a) The reconstruction results for a single-spectral-peak light signal when noise is not concerned and $n_F < n_\lambda = 1601$; (b) the reconstruction results for a single-spectral-peak light signal when noise is concerned, with $S/N = 1000/1$ and $n_F < n_\lambda = 1601$; (c) the results for a broadband light signal when noise is not concerned and $n_F < n_\lambda = 1601$; and (d) the results for a broadband light signal when noise is concerned, with $S/N = 1000/1$ and $n_F < n_\lambda = 1601$.

based on the Tikhonov regularization algorithm can reconstruct the spectrum under noise, but the reconstruction results of the continuous broadband spectrum are poor in detail, and the accuracy of the reconstruction results should be further improved.

Through the analysis of the spectral reconstruction performance of the above three algorithms, the noise tolerance of the spectral reconstruction methods that are commonly used at present is not enough. To increase the accuracy of spectral reconstruction results, it is necessary to propose a method with better noise tolerance.

III. NEW METHOD WITH BETTER NOISE TOLERANCE

For better spectral reconstruction performance, we mainly study the spectral reconstruction method with better noise tolerance in this paper.

At the same time, to reduce the size of the spectrometer, the spectral reconstruction method proposed in this paper should have the ability to reconstruct the spectrum of the source light under the condition that the number of filters is less than the number of spectral bands. The spectral signal is usually sparse, or can be so under some sparsifying basis, so the compressive sensing algorithm introduced above can be used to solve this problem.

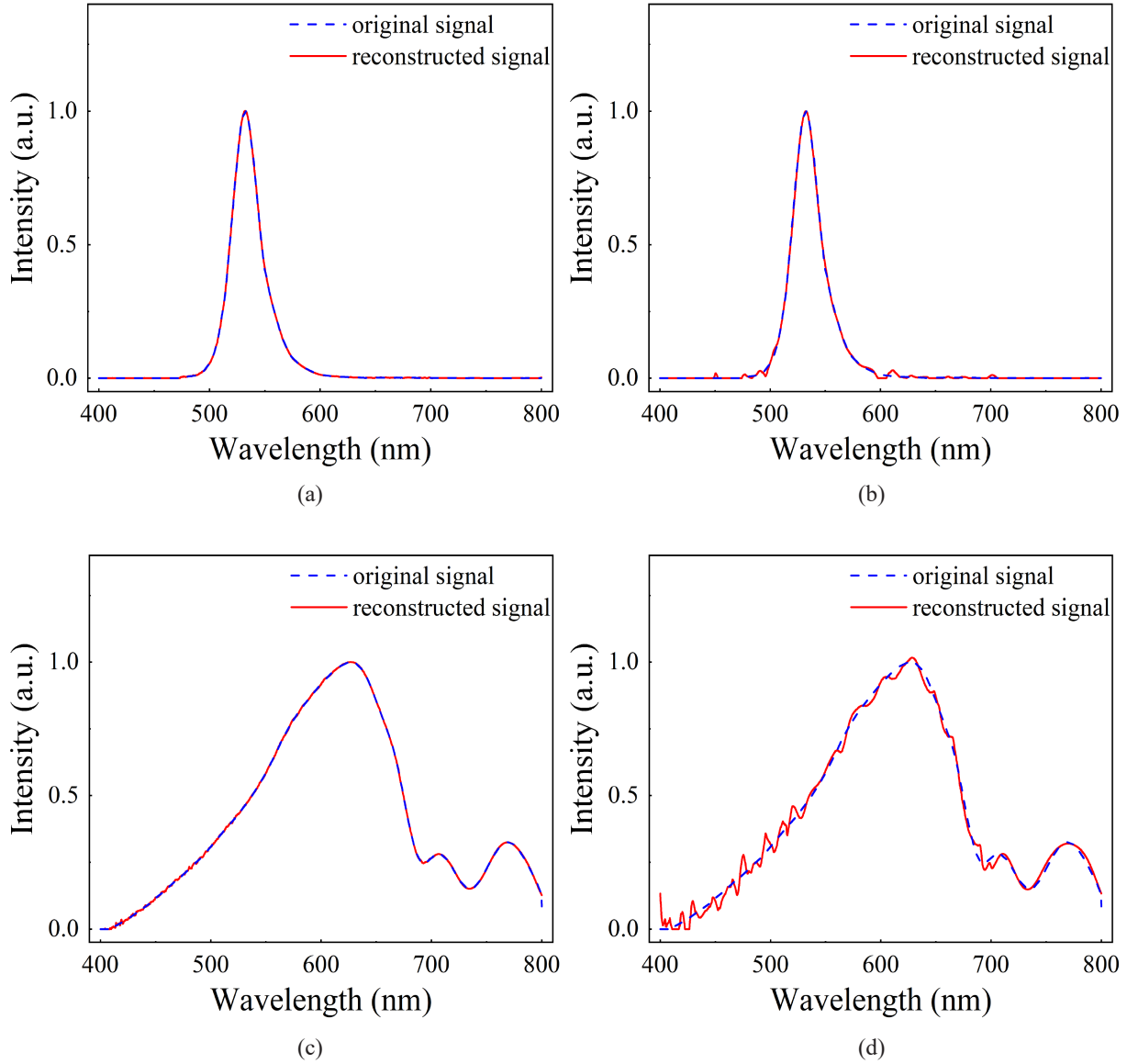


FIG. 7. The effect of system's noise on the Tikhonov regularization algorithm. (a) The reconstruction results for a single-spectral-peak light signal when noise is not concerned and $n_F < n_\lambda = 1601$; (b) the results for a single-spectral-peak light signal when noise is concerned, with $S/N = 1000/1$ and $n_F < n_\lambda = 1601$; (c) the results for a broadband light signal when noise is not concerned and $n_F < n_\lambda = 1601$; and (d) the results for a broadband light signal when noise is concerned, with $S/N = 1000/1$ and $n_F < n_\lambda = 1601$.

The BP algorithm is a typical compressive sensing algorithm; it is proposed to solve Eq. (9), so a spectral reconstruction method based on the BP algorithm can perfectly reconstruct the target spectrum when $n_\lambda > n_F$, but the reconstruction results with this method exhibit very large deviations under noise.

When noise is considered as a factor that affects the reconstruction result, we convert Eq. (3) to the L_1 -regularized least-squares problem:

$$\min_{\theta} \frac{1}{2} \|\mathbf{L}' - \mathbf{\Theta}\theta\|_2^2 + \tau \|\theta\|_1, \quad (11)$$

where τ is a nonnegative real parameter. By solving Eq.

(11), the sparse solution can be obtained and the influence of noise can be reduced as much as possible. After finding the sparse solution $\hat{\theta}$, the reconstruction result $\hat{\Phi}(\lambda)$ can be obtained through $\Phi(\lambda) = \Psi\theta$.

Eq. (11) has attracted great attention for more than three decades, and many signal processing problems where sparseness is sought can be converted to this form. Nowadays, there are several optimization algorithms to solve the problem, such as the basis pursuit de-noising (BPDN) algorithm [29] and the GPSR algorithm [31].

The GPSR algorithm can effectively solve Eq. (11), so it is frequently used to reconstruct two-dimensional image information [17, 32]. However, the algorithm has not been introduced for spectral signal reconstruction. To improve the

spectral reconstruction performance, we propose a spectral reconstruction method based on the GPSR algorithm. The method based on the GPSR algorithm is designed to use fewer filters to achieve high resolution, and has the ability to reconstruct the spectrum from a noisy signal. It proved to be able to achieve satisfactory performance under the noise level of most detectors.

The first step of the GPSR algorithm is to split θ into its positive and negative parts; that is,

$$\theta = \mathbf{u} - \mathbf{v}, \quad \mathbf{u} \geq 0, \mathbf{v} \geq 0. \quad (12)$$

Then Eq. (11) is converted to the bound-constrained quadratic program (BCQP) problem:

$$\min_{\mathbf{z}} \mathbf{c}^T \mathbf{z} + \frac{1}{2} \mathbf{z}^T \mathbf{B} \mathbf{z} \quad \text{subject to } \mathbf{z} \geq 0, \quad (13)$$

where $\mathbf{z} = [\mathbf{u}, \mathbf{v}]^T$, $\mathbf{b} = \Theta^T \mathbf{L}'$, $\mathbf{c} = \mathbf{c}_{l_{2n}} + [-\mathbf{b}, \mathbf{b}]^T$, $\mathbf{l}_n = [1, 1, 1, \dots, 1]^T$ (a vector consisting of n ones), and $\mathbf{B} = [\Theta^T \Theta, -\Theta^T \Theta; -\Theta^T \Theta, \Theta^T \Theta]$. [31] gives two ways to solve Eq. (13): basic-gradient projection for sparse reconstruction (GPSR-Basic), and the Barzilai-Borwein projection for sparse reconstruction (GPSR-BB). Both algorithms have the advantage of good reconstruction effect, easy implementation, and a large range of applications, and their processing speeds are faster than those of many other algorithms [33, 34]. In this paper, we use the GPSR-BB algorithm to reconstruct the spectrum of the original light, with the sparsifying basis selected as the DCT matrix.

The GPSR-BB algorithm can be easily used in MATLAB by downloading the related toolbox [31]. To improve the performance of the method based on the GPSR algo-

rithm, we should choose an appropriate value for τ in Eq. (11). τ is the regularization parameter, and it can balance the tradeoff between the sparsity of the solution $\hat{\theta}$ and residual error; the determination of τ affects the quality of the reconstruction results. In the process of spectral reconstruction, different spectral signals have different sparsity, so it is difficult to choose the regularization parameter τ . The algorithm in [35] provides a way to choose the regularization parameter more effectively, and we apply it to our spectral reconstruction method to achieve better results. By using the algorithm, we can find and trace all solutions $\hat{\theta}(\tau)$ as a function of the regularization parameter τ . In the L_1 -regularized least squares problem, the regularization parameter τ can have some properties [36]: As $\tau \rightarrow 0$, $\|\hat{\theta}\|_1$ is the minimum among all θ that satisfy $\Theta^T(\Theta\theta - \mathbf{L}') = 0$, and as $\tau \rightarrow \infty$, the optimal solution tends to zero and convergence occurs for a finite value of τ , where $\tau \geq \tau_{\max} = \|\mathbf{L}'\|_{\infty}$. Thus we can start the algorithm with $\tau = \|\mathbf{L}'\|_{\infty}$, then generate all solution paths τ as decreases, and the algorithm terminates when τ reaches 0. The best choice for the regularization parameter τ can be found from the solution paths.

We still use the simulated experiments introduced in the previous section to verify the reconstruction performance of the spectral reconstruction method based on the GPSR algorithm. Figure 8 shows the spectral reconstruction results for the single-spectral-peak and broadband light sources by means of this method. The system's noise in the simulations is set to $S/N = 1000/1$. The results show satisfactory quality; the RMSE of the reconstructed signal in Fig. 8(a) is 5.08×10^{-3} , and that in Fig. 8(b) is 7.58×10^{-3} . All of the major features such as crest position and radiance of the original signal are recovered well.

To compare the reconstruction performance of these

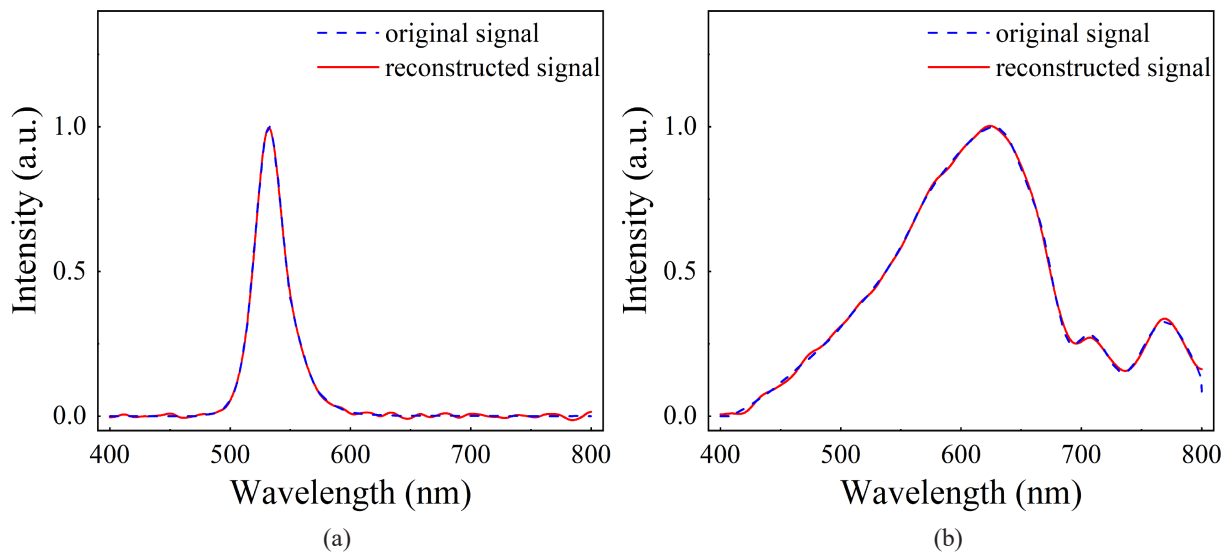


FIG. 8. The simulation results for the spectral reconstruction method based on the GPSR algorithm. (a) The reconstruction results for a single-spectral-peak light signal when noise is concerned, with $S/N = 1000/1$ and $n_F < n_\lambda = 1601$; (b) the results for a broadband light signal when noise is concerned, also with $S/N = 1000/1$ and $n_F < n_\lambda = 1601$.

methods, we add system's noise of varying intensity to the measurements. Tables 1 and 2 show the RMSE of reconstruction results when using different spectral reconstruction methods, for different S/N. The reconstructed light source in Table 1 is the single-spectral-peak light source [Fig. 3(a)], while that in Table 2 is the broadband light source [Fig. 3(b)]. From the simulations we can find that the Tikhonov regularization algorithm, the BPDN algorithm, and the GPSR algorithm can have good performance in recovering the signal when system's noise is concerned, and as S/N increases, the quality of the reconstruction results improves significantly. The results show that the spectral reconstruction method based on the GPSR algorithm can have better performance than that based on the BPDN algorithm. For reconstruction of the single-spectral-peak light source, the methods based on the GPSR and Tikhonov regularization algorithms show similar performance, and when reconstructing the broadband spectrum the GPSR algorithm is significantly better than the Tikhonov regularization algorithm.

To further study the noise tolerance of the method based on the GPSR algorithm, we separately analyze the spectral reconstruction performance under different levels of detector noise and filter noise. We define δ_{Li} as the noise of the i^{th} detector and Δ_{Ti} as the noise of the i^{th} filter, $\Delta_{Ti} = (\delta_{Ti1}, \delta_{Ti2},$

$\delta_{Ti3}, \dots, \delta_{Tin_i})^T$. Therefore, the spectral reconstruction process can be described as

$$L'_i = \sum_{\lambda} [T_i(\lambda) + \Delta_{Ti}] \Phi(\lambda) + \delta_{Li} \quad (i = 1, 2, 3, \dots, n_F). \quad (14)$$

In the simulations, the noise of the i^{th} filter and the detector noise are separately represented by a set of random Gaussian white noise values.

Optoelectronic detectors play an important role in a spectrometer, and according to the characteristics of the detectors, their measurements must contain some amount of noise [37, 38]. These noise can be divided into three main categories: readout noise, dark noise, and photoelectron noise. To analyze the reconstruction performance of the methods, we add detector noise of varying intensity while the filter noise is held at a fixed value. Tables 3 and 4 show the RMSE of the reconstruction results when using different spectral reconstruction methods, for different levels of detector noise.

The filter noise mainly comes from the measurement error of the filter's transmittance, and the drift of the filter's transmittance induced by temperature change. In the simulations, we add filter noise of varying intensity while the detector noise is held at a fixed value. Tables 5 and 6 show

TABLE 1. RMSE of the reconstruction results for different levels of system's noise (single-spectral-peak light source)

S/N	BP	Tikhonov regularization	BPDN	GPSR-BB
300/1	0.95302	0.00907	0.01123	0.00903
500/1	0.97332	0.00630	0.00911	0.00679
1000/1	0.41401	0.00529	0.00649	0.00508

TABLE 2. RMSE of the reconstruction results for different levels of system's noise (broadband light source)

S/N	BP	Tikhonov regularization	BPDN	GPSR-BB
300/1	11.23877	0.04522	0.02894	0.01279
500/1	6.39081	0.03479	0.01521	0.00871
1000/1	3.00997	0.02257	0.01205	0.00758

TABLE 3. RMSE of the reconstruction results for different levels of detector noise (single-spectral-peak light source, intensity of the filter noise is 2.38×10^{-7})

S/N	BP	Tikhonov regularization	BPDN	GPSR-BB
300/1	2.38776	0.00920	0.01297	0.00946
500/1	0.86160	0.00695	0.01018	0.00775
1000/1	0.42955	0.00552	0.00609	0.00375

TABLE 4. RMSE of the reconstruction results for different levels of detector noise (broadband light source, intensity of the filter noise is 2.38×10^{-7})

S/N	BP	Tikhonov regularization	BPDN	GPSR-BB
300/1	8.69321	0.04354	0.02773	0.01864
500/1	5.47973	0.03359	0.01634	0.01262
1000/1	1.68942	0.02388	0.01120	0.00648

TABLE 5. RMSE of the reconstruction results for different levels of filter noise (single-spectral-peak light source, intensity of the detector noise is 0.1034)

S/N	BP	Tikhonov regularization	BPDN	GPSR-BB
300/1	0.46179	0.00569	0.00636	0.00381
500/1	0.41410	0.00549	0.00606	0.00379
1000/1	0.42955	0.00552	0.00609	0.00375

TABLE 6. RMSE of the reconstruction results for different levels of filter noise (broadband light source, intensity of the detector noise is 0.0059)

S/N	BP	Tikhonov regularization	BPDN	GPSR-BB
300/1	1.80551	0.02401	0.01144	0.00691
500/1	1.88617	0.02385	0.01130	0.00653
1000/1	1.68942	0.02388	0.01120	0.00648

the RMSE of the reconstruction results when using different spectral reconstruction methods, for different levels of filter noise.

From the spectral reconstruction results under different levels of detector noise and filter noise, we can draw similar conclusions as for Tables 1 and 2. The method based on the GPSR algorithm can recover the spectrum signal almost perfectly under both detector noise and filter noise, and performs better than other methods.

The above simulations show that the Tikhonov regularization algorithm, the BPDN algorithm, and the GPSR algorithm can all solve the spectral reconstruction problem under noisy conditions, and that among them the GPSR algorithm has the best performance. We briefly analyze the reasons for this phenomenon here. Generally, the Tikhonov regularization algorithm solves the L_2 norm minimization problem [Eq. (10)], and the GPSR algorithm solves the L_1 norm minimization problem [Eq. (11)]. Using the L_1 norm minimization has advantages over its L_2 counterpart. L_1 norm minimization can generate the sparsest solution to the underdetermined problem, and can give robust performance under noisy conditions [39]; this is beneficial for solving the spectral reconstruction problem under noisy conditions. As for the GPSR and BPDN algorithms, both can solve Eq. (11); the difference is the iterative methods they adopt. Using different iterative methods will yield different results. The BPDN algorithm in the simulated experiments uses the Newton method, while the GPSR algorithm uses the Barzilai and Borwein (BB) method [40]. The Newton method is sensitive to initial values, and the local optimal solution obtained may not be the global optimal solution. The reconstruction performance of the BPDN algorithm may be influenced by the disadvantages of the Newton method. Our simulations show that the GPSR-BB algorithm can obtain better spectral reconstruction results under noise.

For further study of the characteristics of the spectral reconstruction method based on the GPSR algorithm, we check the effect of different filter numbers on reconstruction performance. We keep the number of spectral bands n_λ unchanged, and analyze the reconstruction results when the number of filters varies from 30 to 300. We choose the broadband light source [Fig. 3(b)], and the system's noise is set to $S/N = 500/1$. The RMSE of the reconstruction results for different filter numbers is shown in Fig. 9. It is obvious that the accuracy of the reconstruction result improves significantly as the number of filters increases.

We also characterize the spectral separation capacity of the spectral reconstruction method. We site two narrow-band signals close to each other, the spacing between the two peaks being from 10 to 13 nm, and we add noise to the measurements with $S/N = 500/1$. The reconstruction results are shown in Fig. 10. The spectral reconstruction method we propose can recover the signal when the spacing between two peaks is greater than 11 nm; when the spacing is smaller, the two closely positioned spectral signals can hardly be separated.

In addition to designing a spectral reconstruction method with noise tolerance, selecting a suitable sparsifying basis can also be an effective denoising approach. In most situations the original spectrum $\Phi(\lambda)$ is not a sparse or approximately sparse signal, so a sparsifying basis Ψ is also needed in the spectral reconstruction process, and the design of the sparsifying basis will greatly affect the sparse representation of the signal [41, 42]. One way to build the sparsifying basis is by using a traditional transform, such as Fourier transform or wavelet transform [43]. The structure of each of these sparsifying bases is stable, and they are generally applicable to signals whose local features have obvious transformations in time-frequency analysis. As a typical transform, the Fourier transform can describe smooth signals well, and it has a denoising function, yet it can hardly describe piecewise smooth signals. To better describe a signal that is not smooth, the wavelet transform can be used; the basis function group of the wavelet transform can be built through the translation and scaling of a generating function. Most spectra are piecewise smooth, so the wavelet transform will exhibit better performance in sparse representation of a spectral signal.

In addition, when using the discrete wavelet transform (DWT) matrix as the sparsifying basis, the sparse signal $\hat{\theta}$ obtained by the reconstruction algorithm can be processed to significantly reduce the noise introduced by ambient light. As shown in Fig. 11, we add white noise to the original broadband spectrum to simulate the noise introduced by ambient light. The spectral reconstruction in Fig. 11(b) shows that the influence of ambient light on the reconstructed spectrum is basically eliminated.

We also find that by designing a suitable filter structure, the spectral reconstruction performance under noise can be greatly improved. The performance of the spectral reconstruction method is related to the correlation among the different response functions of the filters: Smaller correlation

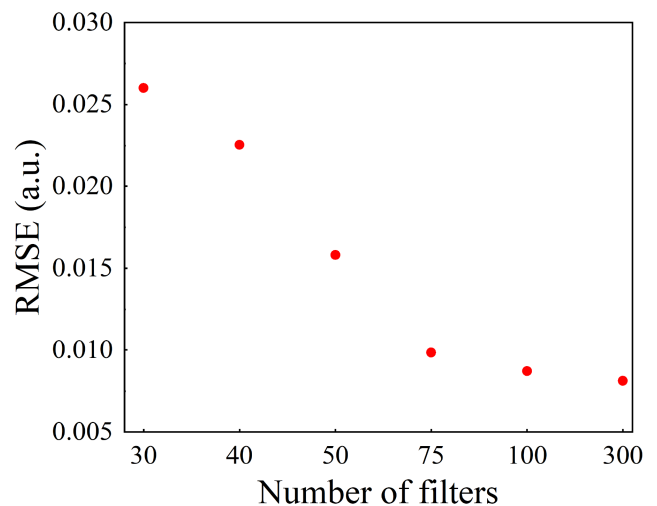


FIG. 9. The effect of the number of filters on reconstruction performance.

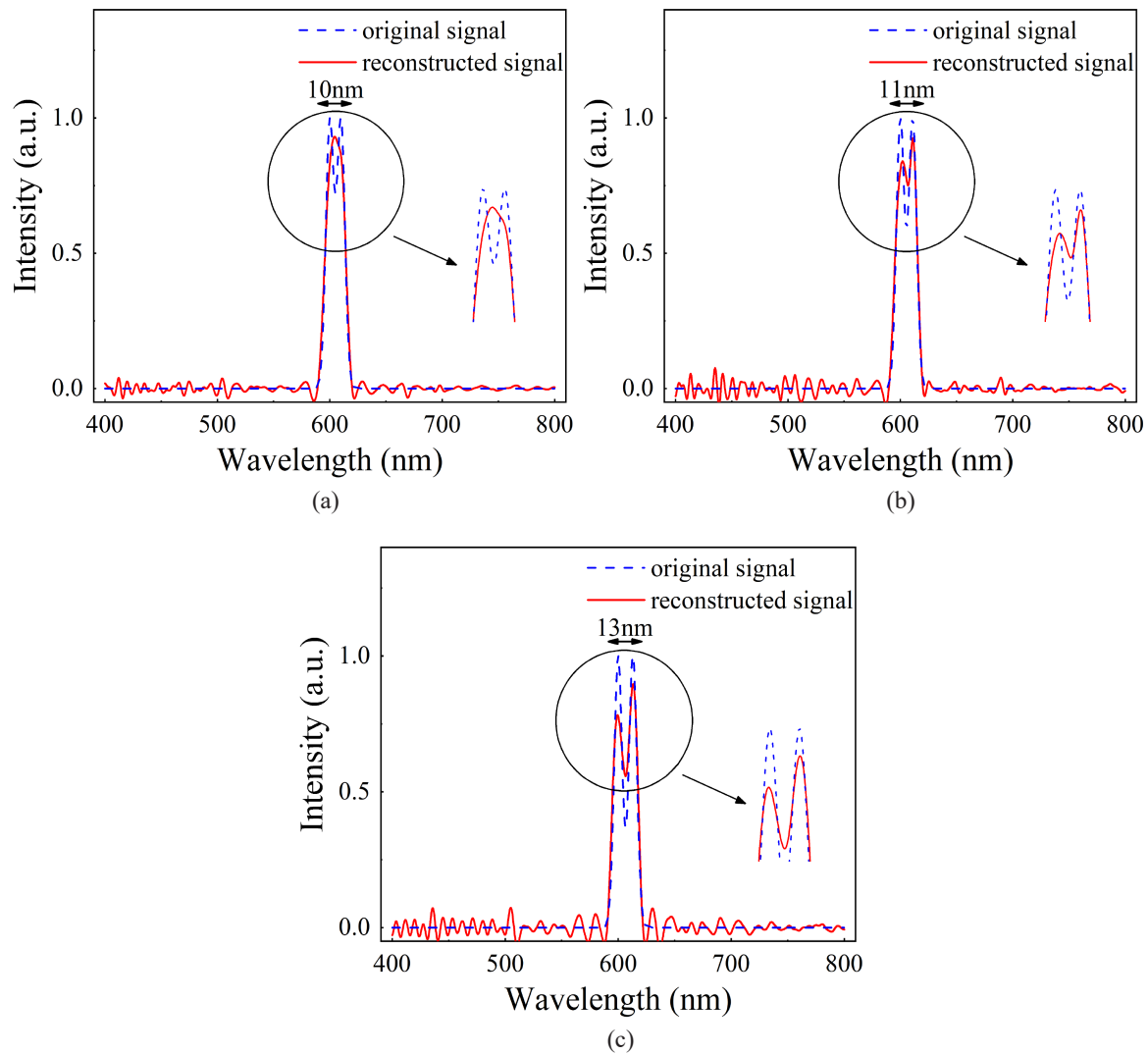


FIG. 10. The spectral separation capacity of the spectral reconstruction method. In (a)–(c) the spacing between two peaks ranges from 10 to 13 nm as indicated.

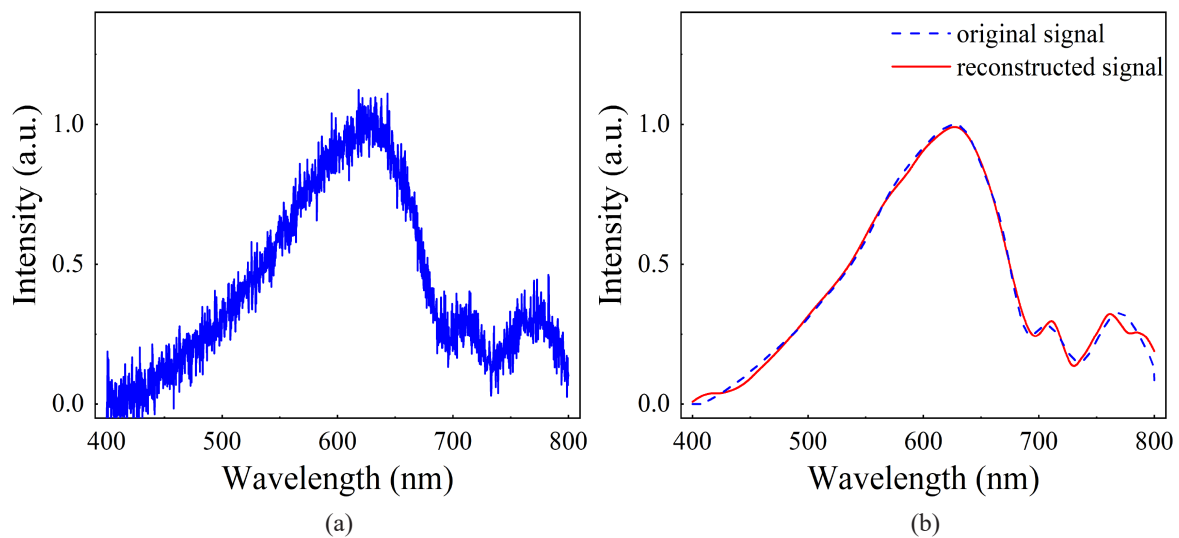


FIG. 11. Denoising effect by using the discrete wavelet transform (DWT) sparsifying basis: (a) original light source with ambient-light noise, and (b) spectral reconstruction results using the DWT sparsifying basis.

TABLE 7. The impact of correlation on reconstruction performance

Average correlation coefficient	RMSE
0.135	0.00650
0.237	0.00871
0.390	0.02839

leads to better reconstruction performance. Here we use the Pearson correlation coefficient to evaluate the correlation among the different response functions of the filter, which can be defined as

$$\rho(\mathbf{T}^k, \mathbf{T}^l) = \frac{\text{cov}(\mathbf{T}^k, \mathbf{T}^l)}{\sigma_{\mathbf{T}^k} \sigma_{\mathbf{T}^l}} = \frac{\sum_{i=1}^{n_2} (T_i^k - \bar{\mathbf{T}}^k)(T_i^l - \bar{\mathbf{T}}^l)}{\sqrt{\sum_{i=1}^{n_2} (T_i^k - \bar{\mathbf{T}}^k)^2} \sqrt{\sum_{i=1}^{n_2} (T_i^l - \bar{\mathbf{T}}^l)^2}}, \quad (15)$$

where $\mathbf{T}^k$ is the response function of the k^{th} filter, $\text{cov}(\mathbf{T}^k, \mathbf{T}^l)$ is the covariance of $\mathbf{T}^k$ and $\mathbf{T}^l$, $\sigma_{\mathbf{T}^k}$ is the standard deviation of $\mathbf{T}^k$, $\bar{\mathbf{T}}^k$ is the average value of $\mathbf{T}^k$, $k, l \in (1, 2, 3, \dots, n_F)$ and $k \neq l$.

To verify the impact of correlation on the reconstruction performance, we compare three sets of filter structures with different average correlation coefficients. We set the system's noise to $S/N = 500/1$, and we choose the broadband light source [Fig. 3(b)]. The RMSE of the reconstruction results for different filter structures is shown in Table 7. As the average correlation coefficient of the filters increases, the reconstruction results become less accurate. Therefore, to obtain more accurate reconstruction under noise, it is necessary to use a set of filters with a smaller correlation.

IV. CONCLUSION

In this paper, the effect of noise on the spectral reconstruction process was analyzed, and we compare the reconstruction performance of different spectral reconstruction methods under noise. To reduce the effect of noise on the filter-based spectrometer, we proposed a spectral reconstruction method based on the GPSR algorithm, and through simulated experiments the superior performance of the proposed method over traditional methods in recovering the spectrum of the source light was proved. The method performs well under noisy conditions, and it can work when the number of filters is less than the number of spectral bands. In addition, we have proposed other denoising approaches, including selecting a suitable sparsifying basis and designing a filter structure with smaller correlation. Such work can be beneficial to make the spectral reconstruction process more useful in applications.

ACKNOWLEDGMENT

This study was supported by the National Natural Sci-

ence Foundation of China under Grant Nos. 61774120 and 61705178.

REFERENCES

1. Y. C. Ha, J. H. Lee, Y. J. Koh, S. K. Lee, and Y. K. Kim, "Development of an ultraviolet Raman spectrometer for standoff detection of chemicals," *Curr. Opt. Photon.* **1**, 247–251 (2017).
2. N. Rajaram, T. J. Aramil, K. Lee, J. S. Reichenberg, T. H. Nguyen, and J. W. Tunnell, "Design and validation of a clinical instrument for spectral diagnosis of cutaneous malignancy," *Appl. Opt.* **49**, 142–152 (2010).
3. A. A. Masoud, K. Koike, M. G. Atwia, M. M. El-Horiny, and K. S. Gmail, "Mapping soil salinity using spectral mixture analysis of landsat 8 OLI images to identify factors influencing salinization in an arid region," *Int. J. Appl. Earth Obs. Geoinf.* **83**, 101944 (2019).
4. C. Kim and T. Ji, "Real-time spectroscopic methods for analysis of organic compounds in water," *Curr. Opt. Photon.* **3**, 336–341 (2019).
5. C. Zhang, G. Cheng, P. Edwards, M.-D. Zhou, S. Zheng, and Z. Liu, "G-Fresnel smartphone spectrometer," *Lab. Chip* **16**, 246–250 (2016).
6. D. R. Lobb, "Imaging spectrometers using concentric optics," *Proc. SPIE* **3118**, 339–347 (1997).
7. H.-A. Lin, H.-Y. Hsu, C.-W. Chang, and C.-S. Huang, "Compact spectrometer system based on a gradient grating period guided-mode resonance filter," *Opt. Express* **24**, 10972–10979 (2016).
8. T. Pügner, J. Knobbe, and H. Gröger, "Near-infrared grating spectrometer for mobile phone applications," *Appl. Spectrosc.* **70**, 734–745 (2016).
9. M. Schardt, P. J. Murr, M. S. Rauscher, A. J. Tremmel, B. R. Wiesent, and A. W. Koch, "Static Fourier transform infrared spectrometer," *Opt. Express* **24**, 7767–7776 (2016).
10. F. Cai, R. Tang, S. Wang, and S. He, "A compact line-detection spectrometer with a Powell lens," *Optik* **155**, 267–272 (2018).
11. X. Ma, M. Li, and J. He, "CMOS-compatible integrated spectrometer based on echelle diffraction grating and MSM photodetector array," *IEEE Photonics J.* **5**, 6600807 (2013).
12. S. Zheng, H. Cai, J. Song, J. Zou, P. Y. Liu, Z. Lin, D.-L. Kwong, and A.-Q. Liu, "A single-chip integrated spectrometer via tunable microring resonator array," *IEEE Photonics J.* **11**, 6602809 (2019).
13. J. Bao and M. G. Bawendi, "A colloidal quantum dot spectrometer," *Nature* **523**, 67–70 (2015).
14. Z. Yang, T. Albrow-Owen, H. Cui, J. Alexander-Webber, F. Gu, X. Wang, T.-C. Wu, M. Zhuge, C. Williams, P. Wang, A. V. Zayats, W. Cai, L. Dai, S. Hofmann, M. Overend, L. Tong, Q. Yang, Z. Sun, and T. Hasan, "Single-nanowire spectrometers," *Science* **365**, 1017–1020 (2019).
15. Z. Wang and Z. Yu, "Spectral analysis based on compressive sensing in nanophotonic structures," *Opt. Express* **22**, 25608–25614 (2014).
16. C. Huang, G. Xia, S. Jin, M. Hu, S. Wu, and J. Xing, "Denoising analysis of compact CCD-based spectrometer," *Optik* **157**,

- 693–706 (2018).
17. H. Yan and Y. Liu, “Improved gradient projection algorithm for deblurred image application,” *J. Phys.: Conf. Ser.* **1575**, 012021 (2020).
 18. X. Zhang, Y. Ren, G. Feng, and Z. Qian, “Compressing encrypted image using compressive sensing,” in *Proc. 2011 Seventh International Conference on Intelligent Information Hiding and Multimedia Signal Processing* (Dalian, China, Oct. 2011), pp. 222–225.
 19. Y. Zuo, J. Zhang, and H. Xi, “Influence of detector noise on infrared images,” *Proc. SPIE* **4548**, 381–386 (2001).
 20. Y. Dai and J. Xu, “The noise analysis and noise reliability indicators of optoelectron coupled devices,” *Solid-State Electron.* **44**, 1495–1500 (2000).
 21. U. Kurokawa, B. I. Choi, and C.-C. Chang, “Filter-based miniature spectrometers: spectrum reconstruction using adaptive regularization,” *IEEE Sens. J.* **11**, 1556–1563 (2011).
 22. C. C. Chang, N.-T. Lin, U. Kurokawa, and B. I. Choi, “Spectrum reconstruction for filter-array spectrum sensor from sparse template selection,” *Opt. Eng.* **50**, 114402 (2011).
 23. C.-C. Chang and H.-N. Lee, “On the estimation of target spectrum for filter-array based spectrometers,” *Opt. Express* **16**, 1056–1061 (2008).
 24. Z. Xu, Z. Wang, M. E. Sullivan, D. J. Brady, S. H. Foulger, and A. Adibi, “Multimodal multiplex spectroscopy using photonic crystals,” *Opt. Express* **11**, 2126–2133 (2003).
 25. Z. Leihong, Y. Xiao, Z. Dawei, and C. Jian, “Research on multiple-image encryption scheme based on Fourier transform and ghost imaging algorithm,” *Curr. Opt. Photon.* **2**, 315–323 (2018).
 26. R.-M. Lan, X.-F. Liu, X.-R. Yao, W.-K. Yu and G.-J. Zhai, “Single-pixel complementary compressive sampling spectrometer,” *Opt. Commun.* **366**, 349–353 (2016).
 27. R. Zhu, G.-S. Li, and Y. Guo, “Block-compressed-sensing-based reconstruction algorithm for ghost imaging,” *OSA Continuum* **2**, 2834–2843 (2019).
 28. Y. C. Eldar and G. Kutyniok, *Compressed Sensing: Theory and Applications* (Cambridge University Press, Cambridge, UK, 2012).
 29. S. S. Chen, D. L. Donoho, and M. A. Saunders, “Atomic decomposition by basis pursuit,” *SIAM Rev.* **43**, 129–159 (2001).
 30. T. Yang, C. Li, Z. Wang, and H. Ho, “An ultra compact spectrometer based on the optical transmission through a micro interferometer array,” *Optik* **124**, 1377–1385 (2013).
 31. M. A. T. Figueiredo, R. D. Nowak, and S. J. Wright, “Gradient projection for sparse reconstruction: application to compressed sensing and other inverse problems,” *IEEE J. Sel. Top. Signal Processing* **1**, 586–597 (2007).
 32. L. Zhang, Y. Zhai, and X. Wang, “Application of Barzilai-Borwein gradient projection for sparse reconstruction algorithm to image reconstruction of electrical capacitance tomography,” *Flow Meas. Instrum.* **65**, 45–51 (2019).
 33. D. Yang, C. Chang, G. Wu, B. Luo, and L. Yin, “Compressive ghost imaging of the moving object using the low-order moments,” *Appl. Sci.* **10**, 7941 (2020).
 34. Z. Wang, X. Ma, R. Chen, G. R. Arce, L. Dong, H.-J. Stock, and Y. Wei, “Comparison of different lithographic source optimization methods based on compressive sensing,” *Proc. SPIE* **11327**, 1132716 (2020).
 35. D. M. Malioutov, M. Cetin, and A. S. Willsky, “Homotopy continuation for sparse signal representation,” *Proc. IEEE International Conference on Acoustics, Speech, and Signal Processing* (Philadelphia, PA, USA, Mar. 2005), Vol. 735, pp. v/733–v/736.
 36. K. Koh, S.-J. Kim, and S. Boyd, “An interior-point method for large-scale ℓ_1 -regularized logistic regression,” *J. Mach. Learn. Res.* **8**, 1519–1555 (2007).
 37. G. Zonios, “Noise and stray light characterization of a compact CCD spectrophotometer used in biomedical applications,” *Appl. Opt.* **49**, 163–169 (2010).
 38. J. J. Davenport, J. Hodgkinson, J. R. Saffell, and R. P. Tatam, “Noise analysis for CCD-based ultraviolet and visible spectrophotometry,” *Appl. Opt.* **54**, 8135–8144 (2015).
 39. J. Oliver, W. Lee, S. Park, and H.-N. Lee, “Improving resolution of miniature spectrometers by exploiting sparse nature of signals,” *Opt. Express* **20**, 2613–2625 (2012).
 40. J. Barzilai and J. M. Borwein, “Two-point step size gradient methods,” *IMA J. Numer. Anal.* **8**, 141–148 (1988).
 41. H. Li, D. Xiang, X. Yang, and X. Zhang, “Compressed sensing method for IGBT high-speed switching time on-line monitoring,” *IEEE Trans. Ind. Electron.* **66**, 3185–3195 (2019).
 42. H.-T. Fang and D.-S. Huang, “Noise reduction in lidar signal based on discrete wavelet transform,” *Opt. Commun.* **233**, 67–76 (2004).
 43. M. Zhang, C. Lu, and C. Liu, “Improved double-threshold denoising method based on the wavelet transform,” *OSA Continuum* **2**, 2328–2342 (2019).